5-4. Correlation – revision

Our data are (x_i, y_i) , $i = 1, 2, \ldots, n$, a random sample from (X, Y)

The distinction between the roles of X and Y in regression analysis is abandoned and we require X to be a **random variable**.

The strength of the association between X and Y (i.e. the strength of the linear relationship between X and Y) is summarised by the **sample correlation coefficient**, r, given as

$$
\begin{aligned}
r &= \frac{S_{xy}}{(S_{xx}S_{yy})^{1/2}} \\
&= \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}
\end{aligned}
$$

Notes:

i. $|r| \leq 1$

ii. When data points fall on a straight line

$$r = +1 \quad \text{or} \quad -1$$

depending on whether the line has a positive or negative slope

iii. In simple linear regression,

$$\begin{aligned} \text{Coefficient of determination} &= \text{Sample correlation between } x \text{ and } y \\ \text{That is,} \qquad R^2 &= r^2 \end{aligned}$$

i.e. $100r^2$ is the percentage of the total variation of the y's which is accounted for by the relationship with x

iv. the value of r gives **no information** about "cause and effect"

v. beware "spurious correlations" – usually the result of the presence of other variables, which you may not even know about.

Inference on $\rho = Corr(X, Y)$

Suppose (X, Y) have a *bivariate normal* distribution with correlation coefficient ρ

Recall definition of population correlation:

$$\rho = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

There are two particularly useful results

(i) Result 1 : Testing for **no association**
Under $H_0 : \rho = 0$,

$$\frac{r\sqrt{n-2}}{\sqrt{1-r^2}} \sim t_{n-2}$$

This enables us to test $H_0 : \rho = 0$ by comparing the computed test statistic to critical values of t_{n-2} distribution.

Equivalent to testing "$b = 0$" in the simple linear regression context.

However, this result cannot be used to test $H_0 : \rho = \rho_0$ for non-zero ρ_0, and cannot be used to find CI for ρ.

(ii) Result 2 : Fisher's transformation of r

$$\text{Let} \quad W = \tanh^{-1}(r) \qquad \left[= \frac{1}{2} \log \left(\frac{1+r}{1-r} \right) \right]$$

Then

$$W \sim N\left(\tanh^{-1}(\rho), \frac{1}{n-3} \right)$$

This is a more general result than in (i). It can be used to test $H_0 : \rho = \rho_0$ for any ρ_0, and to find CIs for ρ.

To find a 95% CI for ρ, first notice that 95% CI for $\tanh^{-1}(\rho)$ is

$$W \pm 1.96 \sqrt{\frac{1}{n-3}}$$

So a 95% CI for ρ is

$$\left(\tanh\left(W - 1.96\sqrt{\frac{1}{n-3}}\right),\ \tanh\left(W + 1.96\sqrt{\frac{1}{n-3}}\right)\right)$$

$$= \left(\tanh\left(\tanh^{-1}(r) - 1.96\sqrt{\frac{1}{n-3}}\right),\ \tanh\left(\tanh^{-1}(r) + 1.96\sqrt{\frac{1}{n-3}}\right)\right)$$

Note: have used here the fact that $\tanh^{-1}(r)$ is a monotonic increasing function for $-1 < r < 1$.

More generally, $100(1-\alpha)\%$ CI for ρ is

$$\left(\tanh\left(\tanh^{-1}(r) - z_{\frac{\alpha}{2}}\sqrt{\frac{1}{n-3}}\right),\ \tanh\left(\tanh^{-1}(r) + z_{\frac{\alpha}{2}}\sqrt{\frac{1}{n-3}}\right)\right)$$